EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

General Equation of SHM

1. The equation of motion of a particle is $x = a\cos(\alpha t)^2$. The motion is

(A) Periodic but not oscillatory.
(B) Periodic and oscillatory.
(C) Oscillatory but not periodic.
(D) Neither periodic nor oscillatory.

PSH084

2. **Statement 1** : Position–time equation of a particle moving along x–axis is $x = 4 + 6\sin\omega t$. Under this situation, motion of particle is not simple harmonic.

and

Statement 2 : $\frac{d^2x}{dt^2}$ for the given equation is not proportional to $-x$.

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.
(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.
(C) Statement–1 is True, Statement–2 is False.
(D) Statement–1 is False, Statement–2 is True.

PSH085

3. A simple harmonic motion having an amplitude A and time period T is represented by the equation: $y = 5\sin\pi(t + 4)$ m. Then the values of A (in m) and T (in sec) are :

(A) $A = 5$; $T = 2$ (B) $A = 10$; $T = 1$ (C) $A = 5$; $T = 1$ (D) $A = 10$; $T = 2$

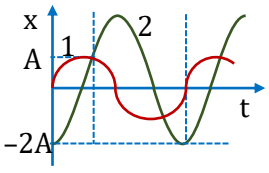
PSH086

4. For a particle acceleration is defined as $\vec{a} = \frac{-5x\hat{i}}{|x|}$ for $x \neq 0$ and $\vec{a} = 0$ for $x = 0$. If the particle is initially at rest at $(a, 0)$, what is period of motion of the particle.

(A) $4\sqrt{2a/5}$ sec. (B) $8\sqrt{2a/5}$ sec. (C) $2\sqrt{2a/5}$ sec. (D) Cannot be determined

PSH088

5. The oscillations represented by curve 1 in the graph are expressed by equation $x = A\sin\omega t$. The equation for the oscillations represented by curve 2 is expressed as:



(A) $x = 2A\sin(\omega t - \pi/2)$ (B) $x = 2A\sin(\omega t + \pi/2)$
(C) $x = -2A\sin(\omega t - \pi/2)$ (D) $x = A\sin(\omega t - \pi/2)$

PSH089

For More Join: @IITJEE_Advanced

6. A particle is performing S.H.M. and at $t = \frac{3T}{4}$, is at position $\frac{\sqrt{3}A}{2}$ and moving towards the origin. Equilibrium position of the particle is at $x = 0$. Then what was the position and direction of particle at $t = 0$?

(A) $-\frac{A}{2}$, away from mean position (B) $\frac{A}{2}$, away from mean position

(C) $\frac{A}{2}$, towards mean position (D) $-\frac{A}{2}$, towards mean position

PSH091

7. The time taken by a particle performing SHM to pass from point A to B where its velocities are same is 2 seconds. After another 2 seconds it returns to B. The time period of oscillation is (in seconds)

(A) 2 (B) 8 (C) 6 (D) 4

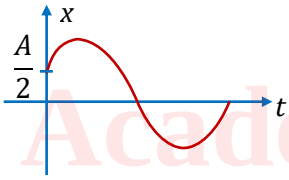
PSH092

8. A particle moves along the x-axis according to : $x = A\ [1 + \sin \omega t]$. What distance does it travel between $t = 0$ and $t = 2.5\pi/\omega$?

(A) $4A$ (B) $6A$ (C) $5A$ (D) None

PSH093

9. Following graph shows a particle performing S.H.M. about mean position $x = 0$. The equation of particle if $t = \frac{T}{4}$ is taken as starting time is (Notations have usual meanings)



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(A) $A\sin\left(\omega t + \frac{2\pi}{3}\right)$ (B) $A\sin\left(\omega t + \frac{\pi}{3}\right)$ (C) $A\sin\left(\omega t + \frac{\pi}{6}\right)$ (D) $A\cos\left(\omega t + \frac{2\pi}{3}\right)$

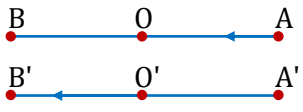
PSH095

10. Two pendulums have time periods T and $5T/4$. They start SHM at the same time from the mean position. After how many oscillations of the smaller pendulum they will be again in the same phase:

(A) 5 (B) 4 (C) 11 (D) 9

PSH097

11. Two particles undergo SHM along parallel lines with the same time period (T) and equal amplitudes. At a particular instant, one particle is at its extreme position while the other is at its mean position. They move in the same direction. They will cross each other after a further time



(A) $T/8$ (B) $3T/8$ (C) $T/6$ (D) $4T/3$

PSH098

12. Two particles are in SHM in a straight line about same equilibrium position. Amplitude A and time period T of both the particles are equal. At time $t = 0$, one particle is at displacement $y_1 = +A$ and the other at $y_2 = -A/2$, and they are approaching towards each other. After what time they cross each other ?

(A) $T/3$ (B) $T/4$ (C) $5T/6$ (D) $T/6$

PSH099

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13. Two particles execute SHM of same amplitude of 20 *cm* with same period along the same line about the same equilibrium position. The maximum distance between the two is 20 *cm*. Their phase difference in radians is :-

(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

PSH101

14. A particle performing SHM is found at its equilibrium at $t = 1$ sec. and it is found to have a speed of 0.25 m/s at $t = 2$ sec. If the period of oscillation is 6 sec. Calculate amplitude of oscillation

(A) $\frac{3}{2\pi}m$ (B) $\frac{3}{4\pi}m$ (C) $\frac{6}{\pi}m$ (D) $\frac{3}{8\pi}$

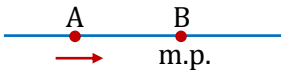
PSH103

15. A particle performs SHM with a period T and amplitude a. The mean velocity of the particle over the time interval during which it travels a distance $a/2$ from the extreme position is

(A) a/T (B) $2a/T$ (C) $3a/T$ (D) $a/2T$

PSH104

16. A particle is oscillating simple harmonically with angular frequency ω and amplitude A. It is at a point (A) at certain instant (shown in figure). At this instant it is moving towards mean position (B). It takes time t to reach mean position (B). If time period of oscillation is T, the average speed between A and B is :-



(A) $\frac{A\sin\omega t}{t}$ (B) $\frac{A\cos\omega t}{t}$ (C) $\frac{A\sin\omega t}{T}$ (D) $\frac{A\cos\omega t}{T}$

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PSH105

17. A particle executes SHM on a straight-line path. The amplitude of oscillation is 2 *cm*. When the displacement of the particle from the mean position is 1 *cm*, the numerical value of magnitude of acceleration is equal to the numerical value of magnitude of velocity. The frequency of SHM (in second^{-1}) is:

(A) $2\pi\sqrt{3}$ (B) $\frac{2\pi}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2\pi}$ (D) $\frac{1}{2\pi\sqrt{3}}$

PSH107

18. A plank with a small block on top of it is under going vertical SHM. Its period is 2 sec. The minimum amplitude at which the block will separate from plank is :

(A) $\frac{10}{\pi^2}$ (B) $\frac{\pi^2}{10}$ (C) $\frac{20}{\pi^2}$ (D) $\frac{\pi}{10}$

PSH108

19. Speed v of a particle moving along a straight line, when it is at a distance x from a fixed point on the line is given by $v^2 = 108 - 9x^2$ (all quantities in S.I. unit). Then

(A) The motion is uniformly accelerated along the straight line.

(B) The magnitude of the acceleration at a distance 3 *cm* from the fixed point is 0.27 m/s^2.

(C) The motion is simple harmonic about $x = \sqrt{12}m$.

(D) The maximum displacement from the fixed point is 4 *cm*.

PSH109

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Energy of Particle in SHM

20. Potential energy of a particle is given as $U(x) = 2x^3 - 9x^2 + 12x$ where U is in joule and x is in metre. If the motion of a particle is S.H.M., then find the potential energy of the particle at mean position :-

(A) $-36\ J$ (B) $4\ J$ (C) $5\ J$ (D) None of these

PSH112

21. If the potential energy of a harmonic oscillator of mass 2 kg on its equilibrium position is 5 joules and the total energy is 9 joules. If the amplitude is one meter then period of the oscillator (in sec) is:

(A) 1.5 (B) 3.14 (C) 6.28 (D) 4.67

PSH113

22. Kinetic energy of a particle executing simple harmonic motion in straight line is pv^2 and potential energy is qx^2, where v is speed at distance x from the mean position. It's time period is given by the expression :-

(A) $2\pi\sqrt{\frac{q}{p}}$ (B) $2\pi\sqrt{\frac{p}{q}}$ (C) $2\pi\sqrt{\frac{q}{p+q}}$ (D) $2\pi\sqrt{\frac{p}{p+q}}$

PSH115

23. Two bodies P and Q of equal mass are suspended from two separate massless springs of force constants k_1 and k_2 respectively. If the maximum velocity of them are equal during their motion, the ratio of amplitude of P to Q is :

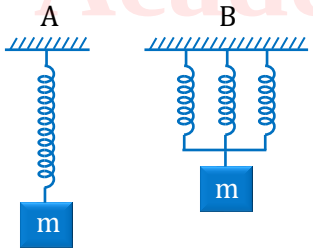
(A) $\frac{k_1}{k_2}$ (B) $\sqrt{\frac{k_2}{k_1}}$ (C) $\frac{k_2}{k_1}$ (D) $\sqrt{\frac{k_1}{k_2}}$

PSH116

Linear SHM

24. The springs in figure A and B are identical but length of spring in A is three times than the length of each spring in B. The ratio of period T_A/T_B is :-

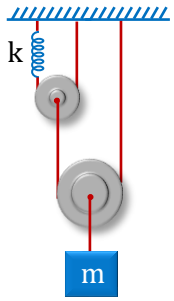
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(A) $\sqrt{3}$ (B) $1/3$ (C) 3 (D) $1/\sqrt{3}$

PSH121

25. What is the period of small oscillations of the block of mass m if the springs are ideal and pulleys are massless ?

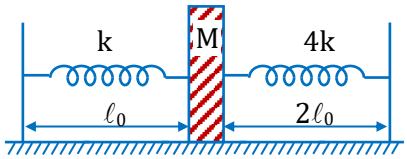


(A) $\frac{\pi}{2}\sqrt{\frac{m}{k}}$ (B) $\frac{\pi}{2}\sqrt{\frac{m}{2k}}$ (C) $\frac{\pi}{2}\sqrt{\frac{2m}{k}}$ (D) $\pi\sqrt{\frac{m}{2k}}$

PSH122

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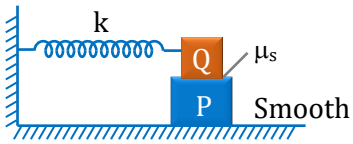
26. A block of mass M is kept in gravity free space and touches the two springs as shown in the figure. Initially springs are in their natural lengths. Now, the block is shifted $(\ell_0/2)$ from the given position in such a way that it compresses a spring and is released. The time-period of oscillation of mass will be



(A) $\frac{\pi}{2}\sqrt{\frac{M}{K}}$ (B) $2\pi\sqrt{\frac{M}{5K}}$ (C) $\frac{3\pi}{2}\sqrt{\frac{M}{K}}$ (D) $\pi\sqrt{\frac{M}{2K}}$

PSH123

27. A block P of mass m is placed on a frictionless horizontal surface. Another block Q of same mass is kept on P and connected to the wall with the help of a spring of spring constant k as shown in the figure. μ_s is the coefficient of friction between P and Q. The blocks move together performing SHM of amplitude A. The maximum value of the friction force between P and Q is :-

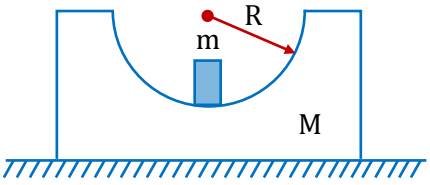


(A) kA (B) $\frac{kA}{2}$ (C) zero (D) $\mu_s mg$

PSH124

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28. A symmetrical block of mass M having a notch of hemispherical shape of radius R is placed on a horizontal surface. A small particle of mass m is placed inside the hemispherical notch and can slide without friction. Find the time period of small oscillation of the particle and the block. The acceleration due to gravity is g.



(A) $2\pi\sqrt{\frac{R}{g\left[1+\frac{m}{M}\right]}}$ (B) $2\pi\sqrt{\frac{R}{g\left[1+\frac{M}{m}\right]}}$ (C) $2\pi\sqrt{\frac{R}{g}\left[1+\frac{m}{M}\right]}$ (D) $2\pi\sqrt{\frac{R}{g}\left[1+\frac{M}{m}\right]}$

PSH141

Angular SHM

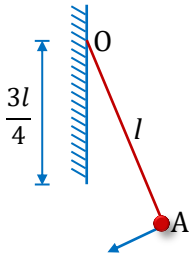
29. A bob is attached to a long, light string. The string is deflected by 3° initially with respect to vertical. The length of the string is 1 m. The value of θ at any time t after the bob released can be approximately written as (Use : $g = \pi^2$)

(A) $3° \cos \pi t$ (B) $3° \sin \pi t$ (C) $3° \sin (\pi t + \frac{\pi}{6})$ (D) $3° \cos (\pi t + \frac{\pi}{6})$

PSH117

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30. A small bob attached to a light inextensible thread of length l has a periodic time T when allowed to vibrate as a simple pendulum. The thread is now suspended from a fixed end O of a vertical rigid rod of length $\frac{3l}{4}$ (as in figure). If now the pendulum performs periodic oscillations in this arrangement, the periodic time will be



(A) $\frac{3T}{4}$ (B) $\frac{T}{2}$ (C) T (D) $2T$

PSH118

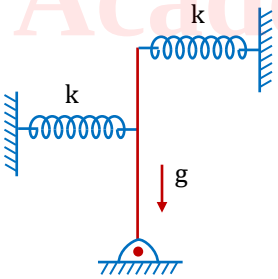
31. A simple pendulum has time period $2s$. The point of suspension is now moved upward according to relation $y = (6t - 3.75\ t^2)m$ where t is in second and y is the vertical displacement in upward direction. The new time period of simple pendulum will be

(A) $2s$ (B) $1s$ (C) $4s$ (D) None of these

PSH119

32. In the figure shown, the spring are connected to the rod at one end and at the midpoint. The rod is hinged at its lower end. Rotational SHM of the rod (Mass m, length L) will occur only if

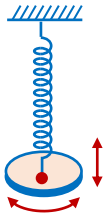
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(A) $k > mg/3L$ (B) $k > 2mg/3L$ (C) $k > 2mg/5L$ (D) $k > 0$

PSH127

33. A solid disk of radius R is suspended from a spring of linear spring constant k and torsional constant c, as shown in figure. In terms of k and c, what value of R will give the same period for the vertical and torsional oscillations of this system?



(A) $\sqrt{\frac{2c}{k}}$ (B) $\sqrt{\frac{c}{2k}}$ (C) $2\sqrt{\frac{c}{k}}$ (D) $\frac{1}{2}\sqrt{\frac{c}{k}}$

PSH130

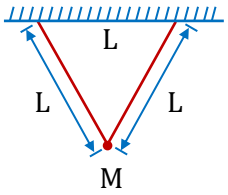
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34. The time period of a bar pendulum when suspended at distances 30 *cm* and 50 *cm* from its centre of gravity comes out to be the same. If the mass of the body is 2*kg*. Find out its moment of inertia about an axis passing through first point.

(A) 0.24 *kg-m*^{2} (B) 0.72 *kg-m*^{2} (C) 0.48 *kg-m*^{2} (D) Data insufficient

PSH131

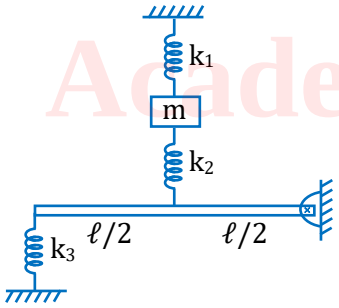
35. A man is swinging on a swing made of 2 ropes of equal length L and in direction perpendicular to the plane of paper. The time period of the small oscillations about the mean position is :-



(A) $2\pi\sqrt{\frac{L}{2g}}$ (B) $2\pi\sqrt{\frac{\sqrt{3}L}{2g}}$ (C) $2\pi\sqrt{\frac{L}{2\sqrt{3}g}}$ (D) $\pi\sqrt{\frac{L}{g}}$

PSH132

36. Find the angular frequency for small oscillation of block of mass m in the arrangement shown in the figure. Neglect the mass of the rod (system is in vertical plane).



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(A) $\omega = \sqrt{\frac{1}{m}\left(k_1 + \frac{4k_2k_3}{k_2+4k_3}\right)}$ (B) $\omega = \sqrt{\frac{1}{m}(k_1+k_2+2k_3)}$

(C) $\omega = \sqrt{\frac{1}{m}\left(\frac{k_1k_2}{k_1+k_3}+4k_3\right)}$ (D) $\omega = \sqrt{\frac{1}{m}\left(k_1+\frac{k_2k_3}{k_2+k_3}\right)}$

PSH142

Superposition Of SHM

37. The amplitude of the vibrating particle due to superposition of two SHMs,

$y_1 = \sin\left(\omega t + \frac{\pi}{3}\right)$ and $y_2 = \sin\omega t$ is

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

PSH135

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38. Two simple harmonic motions $y_1 = A \sin \omega t$ and $y_2 = A \cos \omega t$ are superimposed on a particle of mass m. The total mechanical energy of the particle is:

(A) $\frac{1}{2} m\omega^2 A^2$ (B) $m\omega^2 A^2$ (C) $\frac{1}{4} m\omega^2 A^2$ (D) zero

PSH136

39. The displacement of a particle varies with time according to the relation $y = a \sin \omega t + b \cos \omega t$.

(A) The motion is oscillatory but not S.H.M.

(B) The motion is S.H.M. with amplitude $a + b$.

(C) The motion is S.H.M. with amplitude $a^2 + b^2$.

(D) The motion is S.H.M. with amplitude $\sqrt{a^2 + b^2}$.

PSH137

40. Equations $y = 2A \cos^2 \omega t$ and $y = A (\sin \omega t + \sqrt{3} \cos \omega t)$ represent the motion of two particles.

(A) Only one of these is S.H.M. (B) Ratio of maximum speeds is 2 : 1

(C) Ratio of maximum speeds is 1 : 1 (D) Ratio of maximum accelerations is 1 : 4

PSH138

41. Vertical displacement of a plank with a body of mass 'm' on it is varying according to law $y = \sin \omega t + \sqrt{3} \cos \omega t$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after $t = 0$ are given by: (y is positive vertically upwards)

(A) $\sqrt{\frac{g}{2}}, \frac{\pi}{6}\sqrt{\frac{2}{g}}$ (B) $\frac{g}{\sqrt{2}}, \frac{2}{3}\sqrt{\frac{\pi}{g}}$ (C) $\sqrt{\frac{g}{2}}, \frac{\pi}{3}\sqrt{\frac{2}{g}}$ (D) $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

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PSH139

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. A particle is executing SHM between points-X_m and X_m, as shown in figure-I. The velocity $V(t)$ of the particle is partially graphed and shown in figure-II. Two points A and B corresponding to time t_1 and time t_2 respectively are marked on the $V(t)$ curve.

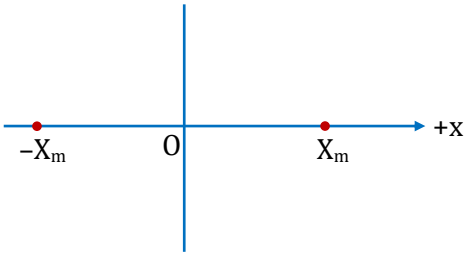


Figure-I

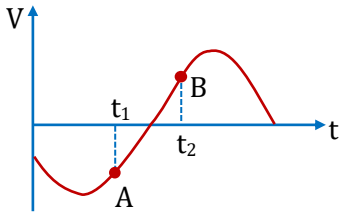


Figure-II

(A) At time t_1, it is going towards X_m.

(B) At time t_1, its speed is decreasing.

(C) At time t_2, its position lies in between $-X_m$ and O.

(D) The phase difference $\Delta\phi$ between points A and B must be expressed as $90° < \Delta\phi < 180°$.

PSH100

2. Two particles are in SHM with same amplitude A and same angular frequency ω. At time $t = 0$, one is at $x = +\frac{A}{2}$ and other is at $x = -\frac{A}{2}$. Both are moving in same direction.

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(A) Phase difference between the two particle is $\frac{\pi}{3}$

(B) Phase difference between the two particle is $\frac{2\pi}{3}$

(C) They will collide after time $t = \frac{\pi}{2\omega}$

(D) They will collide after time $t = \frac{3\pi}{\omega}$

PSH102

3. For a body executing SHM with amplitudes A, time period T, max velocity v_{max} and phase constant zero, which of the following statements are correct?

(A) At $y = (A/2)$, $v > (v_{max}/2)$ (B) $v = (v_{max}/2)$ for $|y| > (A/2)$

(C) For $t = (T/8)$, $y > (A/2)$ (D) For $y = (A/2)$, $t < (T/8)$

PSH106

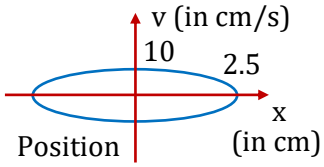
4. A particle is executing SHM with amplitude A, time period T, maximum acceleration a_o and maximum velocity v_0. Its starts from mean position at $t = 0$ and at time t, it has the displacement $A/2$, acceleration a and velocity v then :-

(A) $t = T/12$ (B) $a = a_o/2$ (C) $v = v_o/2$ (D) $t = T/8$

PSH110

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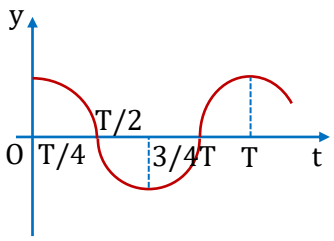
5. The figure shows a graph between velocity and displacement (from mean position) of a particle performing SHM



(A) the time period of the particle is 1.57*s*.
(B) the maximum acceleration will be 40 cm/s^2.
(C) the velocity of particle is $2\sqrt{21}$ cm/s when it is at a distance 1 *cm* from the mean position.
(D) minimum acceleration is 4 cm/s^2.

PSH143

6. The displacement-time graph of a particle executing SHM is shown.

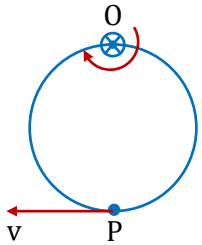


Which of the following statements is/are true?
(A) The velocity is maximum at $t = T/2$.
(B) The acceleration is maximum at $t = T$.
(C) The force is zero at $t = 3T/4$.
(D) The potential energy equals the oscillation energy at $t = T/2$.

Paakets Academy JEE

PSH114

7. A ring of mass *m*, radius *r* can oscillate in a vertical plane about is top most point '*O*' as shown. Axis is normal to the plane of ring. Maximum speed of lowest point '*P*' of ring is *v*.



(A) Time period of SHM for small amplitude is $2\pi\sqrt{\frac{2r}{g}}$.

(B) Acceleration of centre of mass of ring at lowest position is zero.

(C) Hinge force when centre of mass is at lowest point is $\frac{mv^2}{4r}$.

(D) Mechanical energy of oscillation is $\frac{mv^2}{4}$ (when potential energy at mean position is taken as zero)

PSH144

8. A particle moves in *x*-*y* plane according to equation $\vec{S} = (2\hat{i} + \hat{j})\cos\omega t$. The motion of particle is:-
(A) On straight line (B) On ellipse (C) Periodic (D) SHM

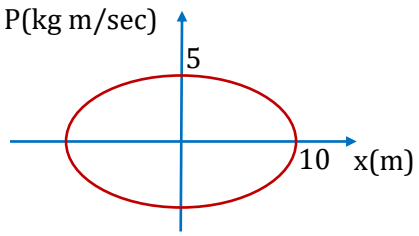
PSH140

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COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 9 AND 10

Momentum (P) displacement (x) graph of a particle of unit mass performing SHM along x-axis is shown in the figure.



9. Angular frequency of the SHM is :

(A) 1 *rad/sec* (B) $\frac{1}{2}$ *rad/sec* (C) 2 *rad/sec* (D) 4 *rad/sec*

PSH145

10. Total energy of oscillation is (potential energy at mean position = 0)

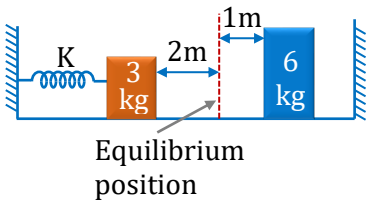
(A) 50 Joule (B) 25 Joule (C) 12.5 Joule (D) 100 Joule

PSH146

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PARAGRAPH FOR QUESTION NO. 11 AND 12

Two blocks of masses 3 *kg* and 6 *kg* rest on a horizontal frictionless surface. The 3 *kg* block is attached to a spring with a force constant K = 900 *N/m* which is compressed 2 *m* initially from its equilibrium position. When 3 *kg* mass is released, it strikes the 6 *kg* mass and the two stick together.



11. The common velocity of the blocks after collision is :-

(A) 10 *m/s* (B) 30 *m/s* (C) 15 *m/s* (D) 2 *m/s*

PSH125

12. The amplitude of resulting oscillation after the collision is :-

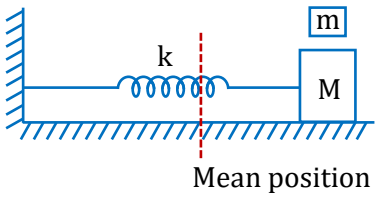
(A) $\frac{1}{\sqrt{2}}m$ (B) $\frac{1}{\sqrt{3}}m$ (C) $\sqrt{2}m$ (D) $\sqrt{3}m$

PSH126

For More Join: @IITJEE_Advanced

PARAGRAPH FOR QUESTION NO. 13 AND 14

A horizontal spring block system executes SHM ($x = A\sin(\omega t + \phi)$) with amplitude $A = 10\ cm$, initial phase $\phi = 0$ and angular frequency ω. The mass of block is $M = 25\ kg$ and there is no friction between the block and the horizontal surface. The spring constant is 2500 N/m. At $t = t_1$ sec [for which $\omega t_1 = 30°$]. A mass $m = 75\ kg$ is gently put on the block. [Assume that collision between the block and the mass is perfectly inelastic and mass m remains stationary *w.r.t.* the block M always].



13. The new angular frequency of the system will be :

(A) 5 *rad/sec* (B) 10 *rad/sec* (C) 15 *rad/sec* (D) 20 *rad/sec*

PSH147

14. The total energy of system after collision at any moment of time is :-

(A) $\frac{175}{32}$ Joule (B) $\frac{88}{15}$ Joule (C) $\frac{96}{25}$ Joule (D) None of these

PSH148

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MATRIX MATCH TYPE QUESTION

15. For a particle under going linear SHM about $x = 0$, find the possible combination. Symbols have their usual meaning.

	List–I		**List–II**
(P)	$(\vec{v}\cdot\vec{a}) > 0$	(1)	Extreme position
(Q)	$v < 0$	(2)	Mean position
(R)	$a < 0$	(3)	$0 < x < A$
(S)	$\vec{v}\times\vec{a} = 0$	(4)	$-A < x < 0$

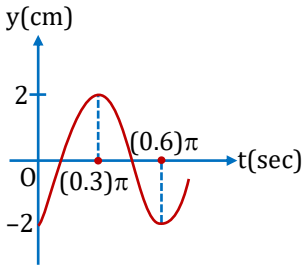
PSH149

For More Join: @IITJEE_Advanced

EXERCISE (S)

General Equation of SHM

1. Part of a simple harmonic motion is graphed in the figure, where y is the displacement from the mean position. The correct equation describing this S.H.M is :-

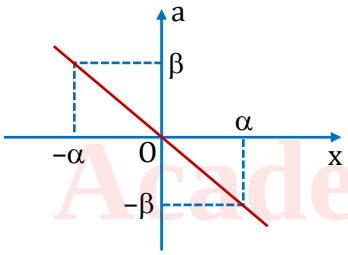


PSH150

2. A body undergoing SHM about the origin has its equation given by $x = 0.2 \cos 5\pi t$. Find its average speed from $t = 0$ to $t = 0.7$ sec.

PSH152

3. The acceleration-displacement ($a - x$) graph of a particle executing simple harmonic motion is shown in the figure. Find the frequency of oscillation.



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PSH153

4. A particle executes simple harmonic motion with an amplitude of 10 cm and time period 6s. At $t = 0$ it is at position $x = 5$ cm going towards positive x-direction. Write the equation for the displacement x at time t. Find the magnitude of the acceleration of the particle at $t = 4s$.

PSH155

Energy of Particle in SHM

5. An object of mass 0.2 kg executes SHM along the x-axis with frequency of $(25/\pi)$ Hz. At the point $x = 0.04m$ the object has KE 0.5 J and PE 0.4 J. The amplitude of oscillation is _____.

PSH156

6. A point particle of mass $0.1kg$ is executing SHM with amplitude of $0.1m$. When the particle passes through the mean position, its $K.E.$ is $8 \times 10^{-3}J$. Obtain the equation of motion of this particle if the initial phase of oscillation is 45°.

PSH157

7. Potential Energy (U) of a body of unit mass moving in a one-dimension conservative force field is given by $U = (x^2 - 4x + 3)$. All units are in S.I.

(i) Find the equilibrium position of the body.

(ii) Show that oscillations of the body about this equilibrium position is simple harmonic motion and find its time period.

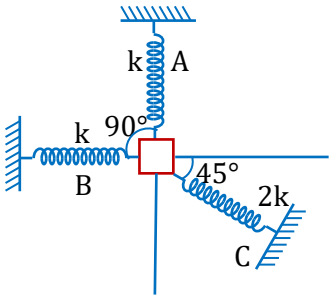
(iii) Find the amplitude of oscillations if speed of the body at equilibrium position is $2\sqrt{6}m/s$.

PSH158

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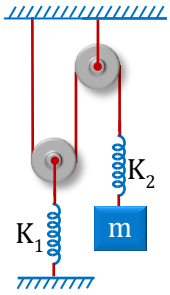
Linear SHM

8. A block of mass m is attached to three springs A, B and C having force constants k, k and $2k$ respectively as shown in figure. Block is slightly pushed against spring C. If the angular frequency of oscillations is $\sqrt{\frac{Nk}{m}}$, then find N. The system is placed on horizontal smooth surface.



PSH128

9. For given spring mass system, If the time period of small oscillations of block about its mean position is $\pi\sqrt{\frac{nm}{K}}$, then find n. Assume ideal conditions. The system is in vertical plane and take $K_1 = 2K$, $K_2 = K$.



PSH129

10. A small body of mass m is fixed to the middle of a stretched string of length 2ℓ. In the equilibrium position the string tension is equal to T_0. Find the angular frequency of small oscillations of the body in the transverse direction. The mass of the string is negligible, the gravitational field is absent. Assume tension in string to be constant.

PSH160

11. A body is in SHM with period T when oscillated from a freely suspended spring. If this spring is cut in two parts of length ratio 1 : 3 and again oscillated from the two parts separately, then the periods are T_1 and T_2 then find T_1/T_2.

PSH161

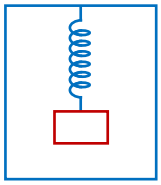
12. A mass M attached to a spring, oscillates with a period of 2 sec. If the mass is increased by 2 kg the period increases by one sec. Find the initial mass M assuming that Hooke's Law is obeyed.

PSH162

13. A spring mass system is hanging from the ceiling of an elevator in equilibrium. Elongation of spring is l. The elevator suddenly starts accelerating downwards with acceleration $g/3$, find:

(a) the frequency and

(b) the amplitude of the resulting SHM.



PSH163

14. A body of mass 1 kg is suspended from a weightless spring having force constant $600 N/m$. Another body of mass 0.5 kg moving vertically upwards hits the suspended body with a velocity of $3.0 m/s$ and get embedded in it. Find the frequency of oscillations and amplitude of motion.

PSH165

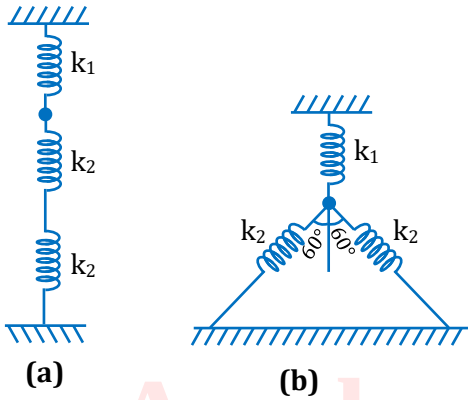
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15. A particle of mass 5×10^{-5} *kg* is placed at the lowest point of a smooth parabola having the equation $20x^2 = y$ (x, y in m). Here y is the vertical height. If it is displaced slightly and it moves such that it is constrained to move along the parabola, the angular frequency of oscillation will be, (in rad/s). If your answer is N fill value $N/4$.

PSH167

16. Mass m is suspended by ideal massless springs in two different ways, indicated by (a) and (b) in the figure. The mass is displaced upwards by a small amount from equilibrium position and is then released resulting in a SHM of the mass in the vertical direction. We denote the oscillation frequencies associated with the two cases (a) and (b) by f_a and f_b respectively. Find $\frac{f_a}{f_b}$.

Given $k_2 = 2k_1$.



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PSH168

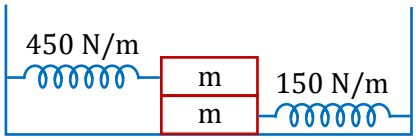
17. A solid sphere of radius R is floating in a liquid of density ρ with half of its volume submerged. If the sphere is slightly pushed and released, it starts performing simple harmonic motion. Find the frequency of these oscillations.

PSH169

18. A spring stores 5 J of energy when stretched by 25 cm. It is kept vertical with the lower end fixed. A block fastened to its other end is made to undergo small oscillations. If the block makes 5 oscillations each second, what is the mass of the block ?

PSH171

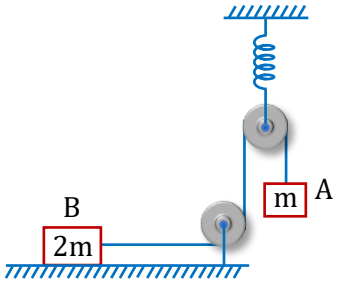
19. When the system shown in the diagram is in equilibrium, the right spring is stretched by 1 cm. The coefficient of static friction between the blocks is 0.3. There is no friction between the bottom block and the supporting surface. The force constants of the springs are 150 N/m and 450 N/m (refer figure). The blocks have equal mass of 2 kg each. Find the maximum amplitude (in cm) of the oscillations of the system shown in the figure that does not allow the top block to slide on the bottom.



PSH173

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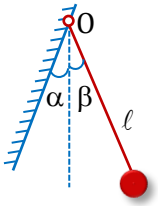
20. In the given figure, spring and pulleys are massless. Block A is performing SHM of amplitude $1m$ and time period π sec. If block B remains at rest, then minimum value of coefficient of friction between block B and surface will be μ. Fill 10μ in OMR sheet. ($g = 10\ m/s^2$)



PSH174

Angular SHM

21. A ball is suspended by a thread of length ℓ at the point O on the wall, forming a small angle α with the vertical as shown in figure. Then the thread with the ball was deviated through a small angle β ($\beta > \alpha$) and set free. Assuming the collision of the ball against the wall to be perfectly elastic, find the oscillation period of such a pendulum.



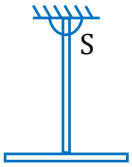
PSH120

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22. A uniform disc of radius r is to be suspended through a small hole made in the disc. Find the minimum possible time period of the disc for small oscillations. What should be the distance of the hole from the centre for it to have minimum time period?

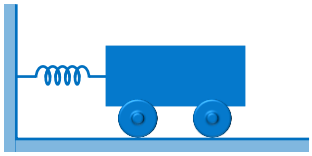
PSH133

23. Two identical rods each of mass m and length L, are rigidly joined and then suspended in a vertical plane so as to oscillate freely about an axis normal to the plane of paper passing through 'S' (point of suspension). Find the time period of such small oscillations.



PSH176

24. A cart consists of a body and four wheels on frictionless axles. The body has a mass m. The wheels are uniform disks of mass M and radius R. The cart rolls, without slipping, back and forth on a horizontal plane under the influence of a spring attached to one end of the cart (figure). The spring constant is k. Taking into account the moment of inertia of the wheels, find a formula for the frequency of the back and forth motion of the cart.



PSH177

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25. (a) Find the time period of oscillations of a torsional pendulum, if the torsional constant of the wire is $K = 10\pi^2 J/rad$. The moment of inertia of rigid body is 10 $kg\ m^2$ about the axis of rotation.

(b) A simple pendulum of length $l = 0.5\ m$ is hanging from ceiling of a car. The car is kept on a horizontal plane. The car starts accelerating on the horizontal road with acceleration of $5\ m/s^2$. Find the time period of oscillations of the pendulum for small amplitudes about the mean position.

PSH178

26. The motion of a simple pendulum is given by

$$\theta = A\cos\left(\sqrt{\frac{g}{\ell}}\,t\right)$$ (symbols have their usual meaning)

(a) Find the tension in the string of this pendulum as a function of time assume that $\theta << \ell$.

(b) At what time is the tension maximum? What is the value of this maximum tension?

PSH179

27. A physical pendulum has the shape of a disk of radius R. The pendulum swings about an axis perpendicular to the plane of the disk and at distance ℓ from the center of the disk.

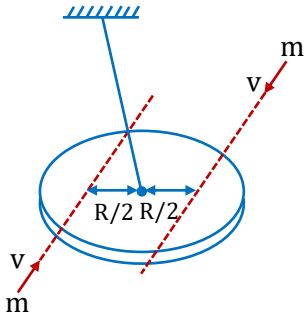
(a) Show that the frequency of the oscillations of this pendulum is $\omega = \sqrt{\dfrac{g\ell}{\frac{1}{2}R^2 + \ell^2}}$

(b) For what value of ℓ is this frequency at a maximum ?

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PSH180

28. A torsional pendulum consists of a uniform disc of mass m and radius R attached to a thin rod of torsional constant C. Two particles of mass m strike with the disc with speed v each and stick to it as shown in figure.



Find the angular amplitude (radian) of oscillation.
[Take $m = 2kg$, $v = 20\ m/s$, $R = 20\ cm$, $C = 80\ Nm/rad$]

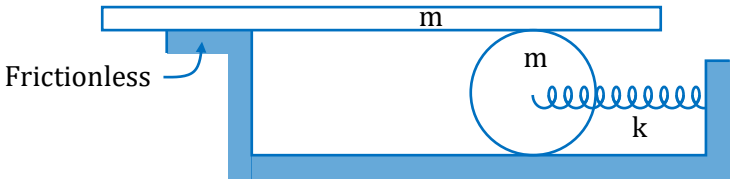
PSH181

29. A rod whose ends are A and B of length 25 cm is hanged in vertical plane. When hanged from point A and point B the time periods calculated are 3 sec and 4 sec respectively. Given the moment of inertia of rod about axis perpendicular to the rod is in ratio 9 : 4 at points A and B. Find the distance of the centre of mass from point A.

PSH182

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30. The axle of a uniform cylinder with mass m and radius R is connected to a spring with spring constant k, as shown in the Fig. A horizontal board with mass m rests on top of the cylinder, and the board also rests on top of a frictionless support near its left end. The system is slightly displaced from equilibrium. There is no slipping between the cylinder and the board, or between the cylinder and the ground. The angular frequency of the oscillatory motion is p. The value of $\frac{11p^2m}{k}$ is:



PSH183

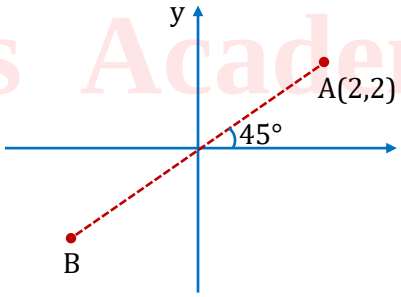
Superposition of SHM

31. A simple harmonic motion is given by $y = 5\ (\sin 3\pi t + \sqrt{3}\cos 3\pi t)$. What is the amplitude (in cm) of motion if y is in m ?

PSH134

32. A particle of mass $m = 2\ kg$ executes SHM in xy-plane between points A and B as shown in figure under action of force $\vec{F} = F_x i + F_y j$. Minimum time taken by particle to move from A to B is 1 sec. At $t = 0$ the particle is at $x = 2$ and $y = 2$. Then find the value of F_x as the function of time.

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PSH184

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EXERCISE - JEE (Main) PYQ

1. A silver atom in a solid oscillates in simple harmonic motion in some direction with a frequency of $10^{12}/sec$. What is the force constant of the bonds connecting one atom with the other ?
(Mole wt. of silver = 108 gm and Avagadro number = 6.02×10^{23} $mole^{-1}$) **[JEE (Main) 2018]**
(1) 2.2 N/m (2) 5.5 N/m (3) 6.4 N/m (4) 7.1 N/m

PSH035

2. A particle executes simple harmonic motion with an amplitude of 5 cm. When the particle is at 4 cm from the mean position, the magnitude of its velocity in SI units is equal to that of its acceleration. Then, its periodic time in seconds is : **[JEE (Main) 2019]**
(1) $\frac{7}{3}\pi$ (2) $\frac{3}{8}\pi$ (3) $\frac{4}{8}\pi$ (4) $\frac{8\pi}{3}$

PSH037

3. A simple pendulum of length 1 m is oscillating with an angular frequency 10 rad/s. The support of the pendulum starts oscillating up and down with a small angular frequency of 1 rad/s and an amplitude of 10^{-2} m. The relative change in the angular frequency of the pendulum is best given by: **[JEE (Main) 2019]**
(1) 10^{-3} rad/s (2) 10^{-1} rad/s (3) 1 rad/s (4) 10^{-5} rad/s

PSH038

4. A pendulum is executing simple harmonic motion and its maximum kinetic energy is K_1. If the length of the pendulum is doubled and it performs simple harmonic motion with the same amplitude as in the first case, its maximum kinetic energy is K_2. Then :- **[JEE (Main) 2019]**
(1) $K_2 = \frac{K_1}{4}$ (2) $K_2 = \frac{K_1}{2}$ (3) $K_2 = 2K_1$ (4) $K_2 = K_1$

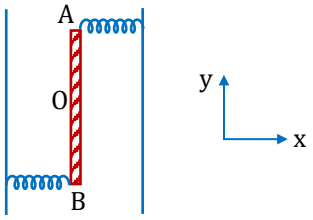
Rankers Academy JEE

PSH039

5. A simple harmonic motion is represented by:
$y = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)$ cm
The amplitude and time period of the motion are: **[JEE (Main) 2019]**
(1) $5cm, \frac{3}{2}s$ (2) $5cm, \frac{2}{3}s$ (3) $10cm, \frac{3}{2}s$ (4) $10cm, \frac{2}{3}s$

PSH041

6. Two light identical springs of spring constant k are attached horizontally at the two ends of a uniform horizontal rod AB of length ℓ and mass m. The rod is pivoted at its centre $'O'$ and can rotate freely in horizontal plane. The other ends of the two springs are fixed to rigid supports as shown in figure. The rod is gently pushed through a small angle and released. The frequency of resulting oscillation is: **[JEE (Main) 2019]**



(1) $\frac{1}{2\pi}\sqrt{\frac{6k}{m}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{2k}{m}}$ (3) $\frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{3k}{m}}$

PSH042

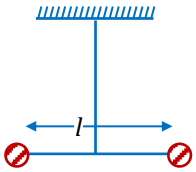
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7. A simple pendulum oscillating in air has period T. The bob of the pendulum is completely immersed in a non-viscous liquid. The density of the liquid is $\frac{1}{16}$ th of the material of the bob. If the bob is inside liquid all the time, its period of oscillation in this liquid is : **[JEE (Main) 2019]**

(1) $4T\sqrt{\frac{1}{15}}$ (2) $2T\sqrt{\frac{1}{10}}$ (3) $4T\sqrt{\frac{1}{14}}$ (4) $2T\sqrt{\frac{1}{14}}$

PSH043

8. Two masses m and $\frac{m}{2}$ are connected at the two ends of a massless rigid rod of length l. The rod is suspended by a thin wire of torsional constant k at the centre of mass of the rod-mass system (see figure). Because of torsional constant k, the restoring torque is $\tau = k\theta$ for angular displacement θ. If the rod is rotated by θ_0 and released so that rod oscillates. The tension in it when it passes through its mean position will be: **[JEE (Main) 2019]**



(1) $\frac{3k\theta_0^2}{l}$ (2) $\frac{k\theta_0^2}{2l}$ (3) $\frac{2k\theta_0^2}{l}$ (4) $\frac{k\theta_0^2}{l}$

PSH044

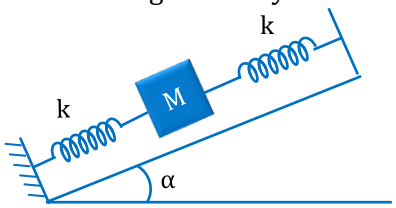
9. A ring is hung on a nail. It can oscillate, without slipping or sliding (i) in its plane with a time period T_1 and, (ii) back and forth in a direction perpendicular to its plane, with a period T_2. The ratio $\frac{T_1}{T_2}$ will be : **[JEE (Main) 2020]**

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(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{\sqrt{2}}{3}$ (3) $\frac{2}{3}$ (4) $\frac{3}{\sqrt{2}}$

PSH045

10. In the given figure, a body of mass M is held between two massless springs, on a smooth inclined plane. The free ends of the springs are attached to firm supports. If each spring has spring constant k, the frequency of oscillation of given body is : **[JEE (Main) 2021]**



(1) $\frac{1}{2\pi}\sqrt{\frac{k}{2M}}$ (2) $\frac{1}{2\pi}\sqrt{\frac{2k}{Mg\sin\alpha}}$ (3) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{k}{Mg\sin\alpha}}$

PSH049

11. $Y = A\sin(\omega t + \phi_0)$ is the displacement-time equation of a SHM. At $t = 0$ the displacement of the particle is $Y = \frac{A}{2}$ and it is moving along negative x-direction. Then the initial phase angle ϕ_0 will be:

[JEE (Main) 2021]

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$ (3) $\frac{5\pi}{6}$ (4) $\frac{2\pi}{3}$

PSH050

For More Join: @IITJEE_Advanced

12. The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is :

[JEE (Main) 2021]

(1) $\sin(\omega t) + \cos(\omega t)$

(2) $\cos(\omega t) + \cos(2\omega t) + \cos(3\omega t)$

(3) $\sin^2(\omega t)$

(4) $3\cos\left(\frac{\pi}{4} - 2\omega t\right)$

PSH051

13. Two simple harmonic motions are represented by the equations $x_1 = 5\ sin\left(2\pi t + \frac{\pi}{4}\right)$ and $x_2 = 5\sqrt{2}\ (\sin 2\pi t + \cos 2\pi t)$. The amplitude of second motion is ..... times the amplitude in first motion.

[JEE (Main) 2021]

PSH054

14. Two simple harmonic motion, are represented by the equations $y_1 = 10 \sin\left(3\pi t + \frac{\pi}{3}\right)$

$y_2 = 5\ (\sin 3\pi t + \sqrt{3}\cos 3\pi t)$

Ratio of amplitude of y_1 to $y_2 = x : 1$. The value of x is _____.

[JEE (Main) 2021]

PSH055

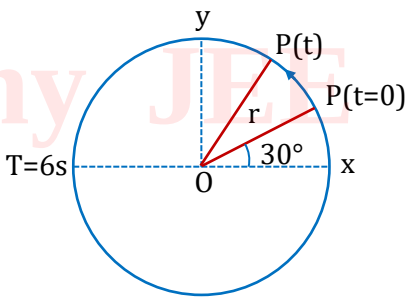
15. The amplitude of a particle executing SHM is 3 cm. The displacement at which its kinetic energy will be 25% more than the potential energy is: _____ cm.

[JEE (Main) 2023]

PSH185

16. For particle P revolving round the centre O with radius of circular path r and angular velocity ω, as shown in below figure, the projection of OP on the x-axis at time t is

[JEE (Main) 2023]



(1) $x(t) = r\cos\left(\omega t + \frac{\pi}{6}\right)$

(2) $x(t) = r\cos(\omega t)$

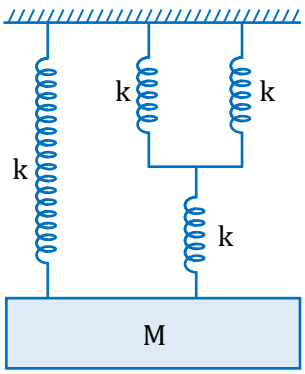
(3) $x(t) = r\sin\left(\omega t + \frac{\pi}{6}\right)$

(4) $x(t) = r\cos\left(\omega t - \frac{\pi}{6}\omega\right)$

PSH187

17. The time period of simple harmonic motion of mass M in the given figure is $\pi\sqrt{\frac{\alpha M}{5K}}$, where the value of α is _____.

[JEE (Main) 2024]



PSH215

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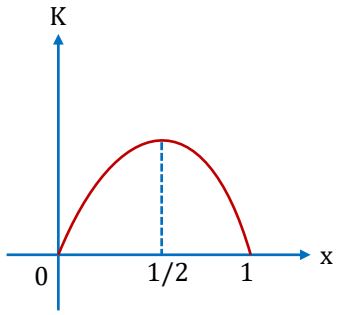
18. When the displacement of a simple harmonic oscillator is one third of its amplitude, the ratio of total energy to the kinetic energy is $\frac{x}{8}$, where x = _____. **[JEE (Main) 2024]**

PSH216

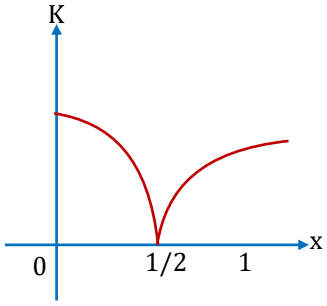
19. A particle oscillates along the x-axis according to the law, $x(t) = x_0 \sin^2\left(\frac{t}{2}\right)$ where $x_0 = 1\ m$. The kinetic energy (K) of the particle as a function of x is correctly represented by the graph.

[JEE (Main) 2025]

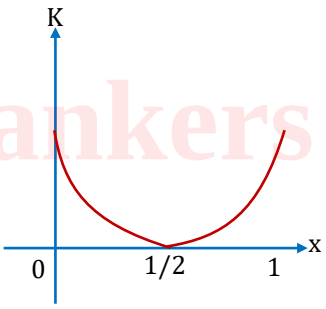
(1)



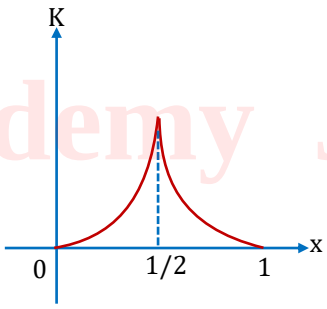
(2)



(3)



(4)



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PSH259

20. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Knowing initial position x_0 and initial momentum p_0 is enough to determine the position and momentum at any time t for a simple harmonic motion with a given angular frequency ω.

Reason (R) : The amplitude and phase can be expressed in terms of x_0 an p_0.

In the light of the above statements, choose the **correct** answer from the options given below:

[JEE (Main) 2025]

(1) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(2) (A) is false but (R) is true.

(3) (A) is true but (R) is false.

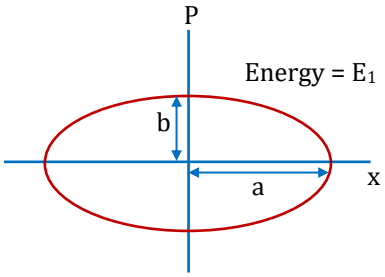
(4) Both (A) and (R) are true and (R) is the correct explanation of (A).

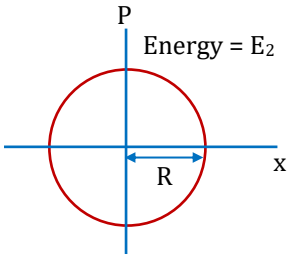
PSH260

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EXERCISE - JEE (Advanced) PYQ

1. Two independent harmonic oscillators of equal mass are oscillating about the origin with angular frequencies ω_1 and ω_2 and have total energies E_1 and E_2, respectively. The variations of their momenta p with positions x are shown in the figures. If $\frac{a}{b}=n^2$ and $\frac{a}{R}=n$, then the correct equation (s) is (are) **[JEE (Advanced) 2015]**





(A) $E_1\omega_1 = E_2\omega_2$ (B) $\frac{\omega_2}{\omega_1}=n^2$ (C) $\omega_1\omega_2 = n^2$ (D) $\frac{E_1}{\omega_1}=\frac{E_2}{\omega_2}$

PSH078

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2. A particle of unit mass is moving along the x-axis under the influence of a force and its total energy is conserved. Four possible forms of the potential energy of the particle are given in column I (a and U_0 are constants). Match the potential energies in column I to the corresponding statement(s) in column-II **[JEE (Advanced) 2015]**

Column-I	**Column-II**
(A) $U_1(x)=\frac{U_0}{2}\left[1-\left(\frac{x}{a}\right)^2\right]^2$	(P) The force acting on the particle is zero at $x = a$.
(B) $U_2(x)=\frac{U_0}{2}\left(\frac{x}{a}\right)^2$	(Q) The force acting on the particle is zero at $x = 0$.
(C) $U_3(x)=\frac{U_0}{2}\left(\frac{x}{a}\right)^2 exp\left[-\left(\frac{x}{a}\right)^2\right]$	(R) The force acting on the particle is zero at $x = -a$.
(D) $U_4(x)=\frac{U_0}{2}\left[\frac{x}{a}-\frac{1}{3}\left(\frac{x}{a}\right)^3\right]$	(S) The particle experiences an attractive force towards $x = 0$ in the region $\lvert x\rvert < a$.
	(T) The particle with total energy $\frac{U_0}{4}$ can oscillate about the point $x = -a$.

PSH079

For More Join: @IITJEE_Advanced

3. A block with mass M is connected by a massless spring with stiffness constant k to a rigid wall and moves without friction on a horizontal surface. The block oscillates with small amplitude A about an equilibrium position x_0. Consider two cases : (i) when the block is at x_0; and (ii) when the block is at $x = x_0 + A$. In both the cases, a particle with mass m ($<M$) is softly placed on the block after which they stick to each other. Which of the following statement(s) is(are) true about the motion after the mass m is placed on the mass M? **[JEE (Advanced) 2016]**

(A) The amplitude of oscillation in the first case changes by a factor of $\sqrt{\frac{M}{m+M}}$, whereas in the second case it remains unchanged

(B) The final time period of oscillation in both the cases is same

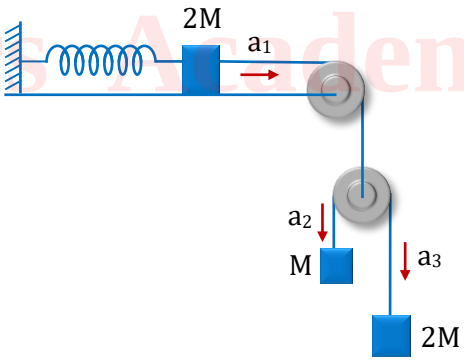
(C) The total energy decreases in both the cases

(D) The instantaneous speed at x_0 of the combined masses decreases in both the cases.

PSH080

4. A block of mass $2M$ is attached to a massless spring with spring-constant k. This block is connected to two other blocks of masses M and $2M$ using two massless pulleys and strings. The accelerations of the blocks are a_1, a_2 and a_3 as shown in figure. The system is released from rest with the spring in its unstretched state. The maximum extension of the spring is x_0. Which of the following option(s) is/are correct ? [g is the acceleration due to gravity. Neglect friction] **[JEE (Advanced) 2019]**

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(A) $x_0 = \frac{4Mg}{k}$

(B) When spring achieves an extension of $\frac{x_0}{2}$ for the first time, the speed of the block connected to the spring is $3g\sqrt{\frac{M}{5k}}$

(C) $a_2 - a_1 = a_1 - a_3$

(D) At an extension of $\frac{x_0}{4}$ of the spring, the magnitude of acceleration of the block connected to the spring is $\frac{3g}{10}$

PSH081

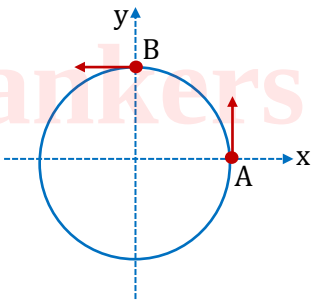
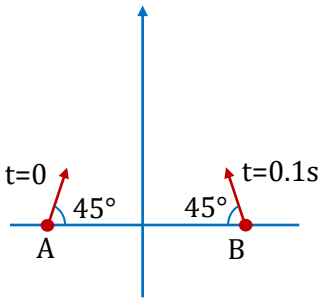
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5. On a frictionless horizontal plane, a bob of mass $m = 0.1\ kg$ is attached to a spring with natural length $l_0 = 0.1\ m$. The spring constant is $k_1 = 0.009\ Nm^{-1}$ when the length of the spring $l > l_0$ and is $k_2 = 0.016\ Nm^{-1}$ when $l < l_0$. Initially the bob is released from $l = 0.15\ m$. Assume that Hooke's law remains valid throughout the motion. If the time period of the full oscillation is $T = (n\pi)\ s$, then the integer closest to n is _____. **[JEE (Advanced) 2021]**

PSH082

6. List I describes four systems, each with two particles A and B in relative motion as shown in figure. List II gives possible magnitudes of the relative velocities (in ms^{-1}) at time $t = \frac{\pi}{3} s$.

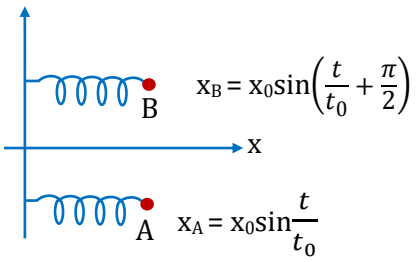
[JEE (Advanced) 2022]

List-I		**List-II**	
(I)	A and B are moving on a horizontal circle of radius 1 m with uniform angular speed $\omega = 1\ rad\ s^{-1}$. The initial angular positions of A and B at time $t = 0$ are $\theta = 0$ and $\theta = \frac{\pi}{2}$ respectively. 	(P)	$\frac{\sqrt{3}+1}{2}$
(II)	Projectiles A and B are fired (in the same vertical plane) at $t = 0$ and $t = 0.1$ s respectively, with the same speed $v = \frac{5\pi}{\sqrt{2}}\ m\ s^{-1}$ and at 45° from the horizontal plane. The initial separation between A and B is large enough so that they do not collide, ($g = 10\ m\ s^{-2}$). 	(Q)	$\frac{\left(\sqrt{3}-1\right)}{\sqrt{2}}$

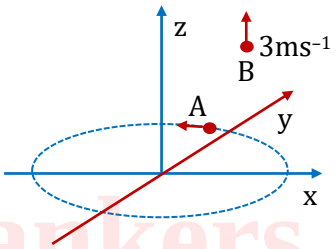
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(III) Two harmonic oscillators A and B moving in the x direction according to $x_A = x_0 \sin\frac{t}{t_0}$ and $x_B = x_0 \sin\left(\frac{t}{t_0}+\frac{\pi}{2}\right)$ respectively, starting from $t = 0$. Take $x_0 = 1\ m$, $t_0 = 1$ s. (R) $\sqrt{10}$



(IV) Particle A is rotating in a horizontal circular path of radius 1 m on the xy plane, with constant angular speed $\omega = 1\ rad\ s^{-1}$. Particle B is moving up at a constant speed 3 $m\ s^{-1}$ in the vertical direction as shown in the figure. (Ignore gravity.) (S) $\sqrt{2}$



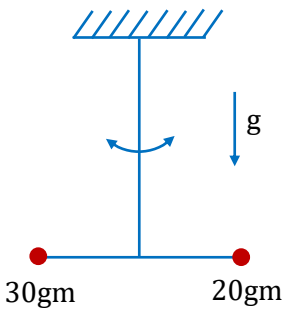
(T) $\sqrt{25\pi^2+1}$

Which one of the following options is correct?

(A) I → R, II → T, III → P, IV → S
(B) I → S, II → P, III → Q, IV → R
(C) I → S, II → T, III → P, IV → R
(D) I → T, II → P, III → R, IV → S

PSH083

7. Two point-like objects of masses 20 gm and 30 gm are fixed at the two ends of a rigid massless rod of length 10 cm. This system is suspended vertically from a rigid ceiling using a thin wire attached to its center of mass, as shown in the figure. The resulting torsional pendulum undergoes small oscillations. The torsional constant of the wire is $1.2 \times 10^{-8}\ N\ m\ rad^{-1}$. The angular frequency of the oscillations is $n \times 10^{-3}\ rad\ s^{-1}$. The value of n is _____. **[JEE (Advanced) 2023]**

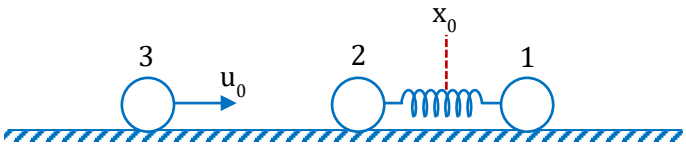


PSH188

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PARAGRAPH FOR Q.NO. 8 and 9

Two particles, 1 and 2, each of mass m, are connected by a massless spring, and are on a horizontal frictionless plane, as shown in the figure. Initially, the two particles, with their center of mass at x_0, are oscillating with amplitude a and angular frequency ω. Thus, their positions at time t are given by $x_1(t) = (x_0 + d) + a \sin \omega t$ and $x_2(t) = (x_0 - d) - a \sin \omega t$, respectively, where $d > 2a$. Particle 3 of mass m moves towards this system with speed $u_0 = a\omega/2$, and undergoes instantaneous elastic collision with particle 2, at time t_0. Finally, particles 1 and 2 acquire a center of mass speed v_{cm} and oscillate with amplitude b and the same angular frequency ω.



[JEE (Advanced) 2024]

8. If the collision occurs at time $t_0 = 0$, the value of $v_{cm}/(a\omega)$ will be _____.

PSH217

9. If the collision occurs at time $t_0 = \pi/(2\omega)$, then the value of $4b^2/a^2$ will be _____.

PSH218

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	A	A	A	A	B	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	C	A	C	A	C	A	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	B	C	A	C	B	A	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	C	C	A	C	B	A	C	B	D	C
Que.	41									
Ans.	A									

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	BC	AC	ABCD	AB	ABC	BCD	AD	ACD	B	C
Que.	11	12	13	14	15					
Ans.	A	C	A	A	(P-3,4), (Q-2,3,4), (R-1,3), (S-1,2,3,4)					

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EXERCISE (S)

1. $y = 2\sin\left(\frac{10}{3}t - \frac{\pi}{2}\right)$ **2.** $2\ m/s$ **3.** $\frac{1}{2\pi}\sqrt{\frac{\beta}{\alpha}}$

4. $x = (10\ cm)\sin\left[\left(\frac{\pi}{3}s^{-1}\right)t + \frac{\pi}{6}\right]$, $\frac{10}{9}\pi^2 \approx 11\ cm/s^2$ **5.** $0.06m$

6. $y = 0.1\sin(4t + \pi/4)$ **7.** (i) $x_0 = 2m$; (ii) $T = \sqrt{2}\,\pi$ sec.; (iii) $2\sqrt{3}$

8. 3 **9.** 12 **10.** $\omega = \sqrt{\frac{2T_0}{m\ell}}$

11. $1/\sqrt{3}$ **12.** $M = 1.6\ kg$

13. (a) $\frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$, (b) $\frac{L}{3}$ **14.** $10/\pi\ Hz$, $\frac{5\sqrt{37}}{6}\ cm$ **15.** 5

16. 1 **17.** $f = \frac{1}{2\pi}\sqrt{\frac{3g}{2R}}$ **18.** $0.16\ kg$

19. 3 **20.** 7

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21. $T = 2\sqrt{\ell / g}$ $[\pi/2 + \sin^{-1}(\alpha/\beta)]$ **22.** $2\pi\sqrt{\frac{r\sqrt{2}}{g}}, r/\sqrt{2}$ **23.** $2\pi\sqrt{\frac{17L}{18g}}$

24. $\omega = \sqrt{\frac{k}{(m+6M)}}$ **25.** (a) 2 sec, (b) $T = \frac{2}{5^{1/4}}$ sec

26. (a) $T = mg + mgA^2 \sin^2 \sqrt{\frac{g}{\ell}}\, t$ (b) $t = (2n+1)\frac{\pi}{2}\sqrt{\frac{\ell}{g}}$; $n \in I$ $T_{max} = mg + mgA^2$

27. (a) $\omega = \sqrt{\frac{g\ell}{\frac{1}{2}R^2 + \ell^2}}$ (b) $\ell = \frac{R}{\sqrt{2}}$ **28.** 2 **29.** 20

30. 2 **31.** 1000 **32.** $-4\pi^2 \cos \pi t$

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	1	3	4	1	1	4	1	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	2	1	2	1	12	9	1	4

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6
Ans.	BD	(A-P,Q,R,T), (B-Q,S), (C-P,Q,R,S), (D-P,R,T)	ABD	C	6	C

Que.	7	8	9
Ans.	10	0.75	4.25

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